

Does Sound Affect Typing Speed?

Investigating the Effect of Auditory Conditions on Typing Speed

STAT 425 – Experimental Design

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Introduction

Many students and professionals listen to music or background audio while working at a computer. Whether this helps or hurts their performance is a common question. Does what you are listening to affect your performance when typing? That is the main question we want to answer with this analysis. To do this we tested it ourselves using a one minute typing test, taking data over the course of two weeks. We recorded who tested the data of our group and what we listened to of four options; Upbeat music, lo-fi music, white noise, and silence. Through this report we will talk about which factors affect performance and which factor levels had the best result at improving your typing speed.

Introducing the Data

Data were collected by three group members who served as both participants and testers: Manreet Kaur Jhattu, Lachlan Findlay, and Zoya Malik. Each tester completed typing trials under four distinct auditory conditions: Upbeat music, Lo-Fi music, White Noise, and Silence. For each condition, two replications were completed and each replication session consisted of three separate one-minute typing trials. This yields $3 \text{ testers} \times 4 \text{ conditions} \times 2 \text{ replications} \times 3 \text{ trials} = 72 \text{ total observations}$, with words per minute (WPM) as the response variable.

A Completely Randomized Block Design (CRBD) was chosen because the three testers are expected to differ systematically in baseline typing speed. By treating tester as a blocking factor, we isolate and remove this between-tester variability from the error term, which improves the power of the test on our factor of interest: auditory condition. The four conditions were selected to span a spectrum from complete silence to highly stimulating vocal music, enabling comparisons between ambient-style audio (Lo-Fi and White Noise), energetic music (Upbeat), and a no-audio baseline (Silence).

Each trial consisted of a standardised one-minute typing test administered through an online typing platform. The order of auditory conditions was randomised for each tester to mitigate carry-over and fatigue effects, and within each session the assigned playlist or audio was played continuously through headphones. Data were collected between March 18th and March 28th, 2026. Prior to analysis, we expected that Upbeat music—with its lyrics and faster tempo—would be the most disruptive to typing performance, while Silence, Lo-Fi, and White Noise would produce broadly similar and higher WPM scores.

To get a look at the data, we look at some summary statistics:

Music <chr>	min <dbl>	Q1 <dbl>	median <dbl>	Q3 <dbl>	max <dbl>	mean <dbl>	sd <dbl>	n <int>	missing <int>
Lo-Fi	54	63.25	67.0	69.5	72	65.66667	4.910972	18	0
Silence	58	63.25	68.5	71.0	74	66.83333	5.382433	18	0
Upbeat	50	60.25	62.5	64.0	71	61.66667	4.862824	18	0
White Noise	55	61.50	65.0	67.0	71	64.05556	4.398826	18	0

Figure 1: WPM broken down by music condition

In Figure 1, we can see a general trend with the mean Words per Minute (WPM) of silence being the highest and best performing and upbeat being the worst performing.

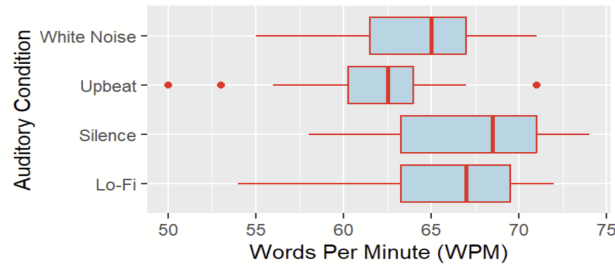


Figure 2: Boxplot of Typing Speed Across Auditory Conditions

From Figure 2, we can see that upbeat is the only treatment with outliers. Upbeat also has the lowest median WPM.

Analysis

Before we begin our analysis, we first need to make a model for our data. In this instance we have the response variable of words per minute, the fixed treatment factor of what was listened to during the tests, and the fixed blocking factor of which tester did each test. From that we get the following model.

$$X_{ij} = \mu + \tau_i + \beta_j + e_{ij}$$

In our model each variable represents the following

μ = the true mean of words per minute.

τ_i = the effect of the i -th level of our Music factor, $i = 1, 2, 3, 4$.

β_j = the effect of the j -th level of our Tester factor, $j = 1, 2, 3$.

e_{ij} = the residuals of our model.

Since we now have our model, we can continue to our analysis. Since we are attempting to determine if the words per minute is affected by the different levels of our factors, we need to use a two way analysis of variance. To do a Completely Randomized Block Design, we need to verify four conditions:

1. The residual terms are normally distributed

2. The variation of the response variable is the same across all levels of treatments and blocking factors
3. Independence
4. The model is additive (no significant interaction)

First, to visualize the normality of our data, we create a plot of the residuals.

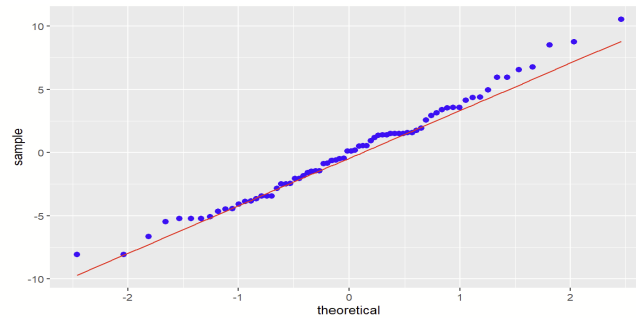


Figure 3: Normal Probability Plot

Based on figure 3, we can see that the data appears normally distributed. We can confirm this result with the Shapiro-Wilk test on residuals. The test produced $W = 0.9867$ and an associated p-value of $p = 0.6536$. There is no evidence against normality with a significance level of 0.05.

Secondly, to determine if there is equal variance, we look at a residuals vs fitted values plot.

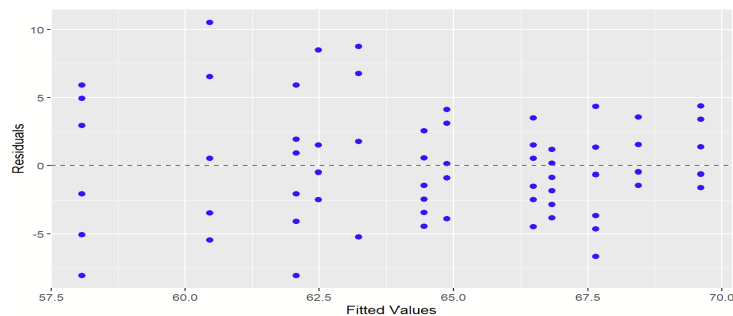


Figure 4: Residuals Plot

In figure 4, we can see the plot looks quite ambiguous which pushed us to run a Levene's Test for equal variance. This test provided us with an f statistic of 0.4332 and a p-value of 0.7299 which allows us to conclude that the variance is equal across the four conditions.

Our third condition is for independence. The trials were randomized across all the conditions and each playlist was shuffled within each session. Each trial consisted of independent one-minute tests.

The last condition is for additivity. We ran an interaction plot below.

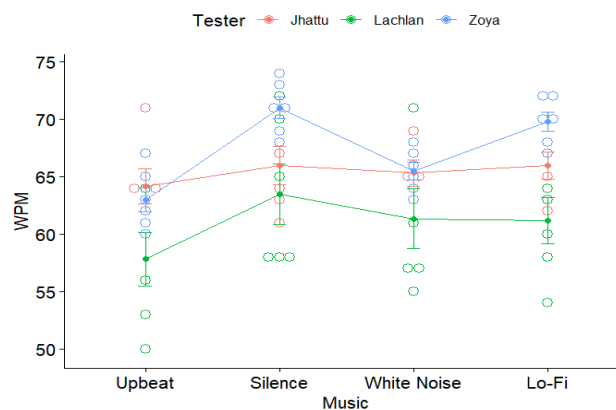


Figure 5: Interaction Plot of Auditory Condition x Tester

In figure 5, we noticed that all three testers have a similar pattern where upbeat produces the lowest WPM. There are individual differences however, lines cross only once. There is no significant interaction between audio condition and tester which is why there is no need for an interaction term in the model.

From above, we have verified that all four of the conditions for CRBD are satisfied. To test if the differences in figure 1 are statistically significant, we run ANOVA.

ANOVA

- We test the auditory condition effect with our null hypothesis of H_0 : the mean WPM is equal for all four auditory conditions. Our alternative hypothesis is H_a : at least one mean differs.
- We test the tester effect with our null hypothesis of H_0 : no tester effect on WPM. Our alternative hypothesis is H_a : at least one tester differs.

We run our ANOVA below:

```

      Df Sum Sq Mean Sq F value    Pr(>F)
Music      3   270.3    90.11    5.303   0.00246 **
Tester      2   511.9   255.93   15.060 0.00000409 ***
Residuals  66  1121.6    16.99
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Figure 6: CRBD ANOVA

The music factor has an f value of 5.303 with associated p-value of 0.00246. The p-value is less than 0.05 and therefore, we reject the null hypothesis. We can conclude that there is evidence here to suggest that audio has a significant effect on typing speed.

The tester block has an f value of 15.06 and p-value of 0.00000409 which is much less than the standard significance level of 0.05. We likewise reject the null hypothesis and conclude that the tester should be treated as a block. This confirms that individual differences in baseline typing speed across the three testers are significant. This validates our decision to use a CRBD: blocking by tester effectively controls for between-tester variability and improves the precision of our treatment comparisons

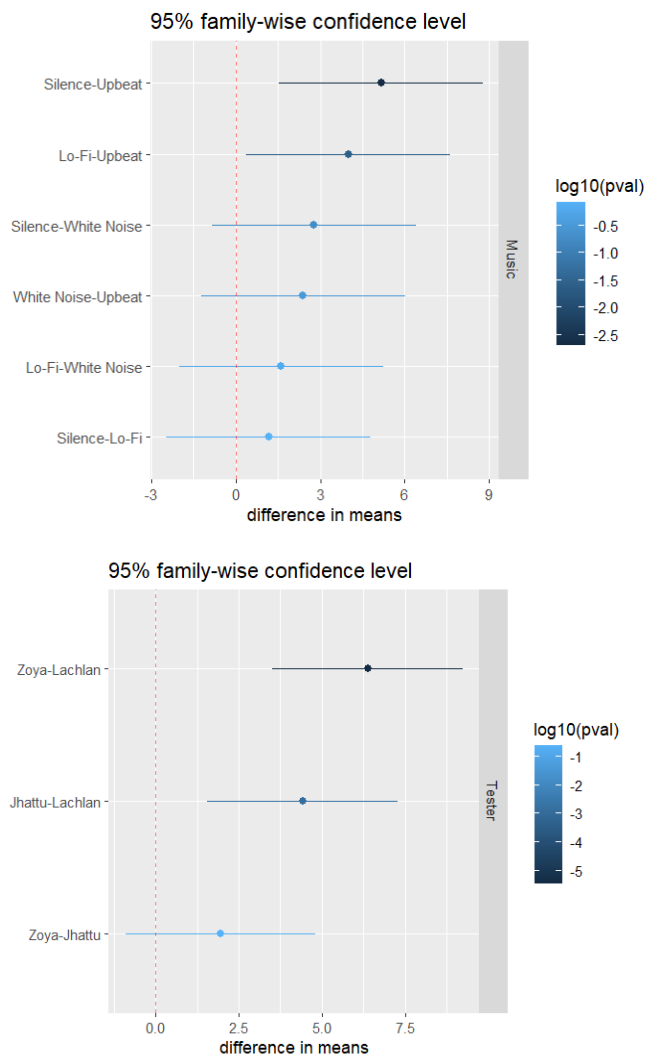
Tukey HSD

From our ANOVA analysis above, we know that the mean auditory conditions differ. We want to

know which one(s) differ and affect productivity in typing speed. To do this, we run a Tukey HSD

test.

Figures 7 and 8 display the Tukey HSD pairwise comparisons and confidence interval plot for the Music factor.



Figures 7 and 8: Tukey HSD Output Table and Confidence Interval Plot for Auditory Condition

At the $\alpha = 0.05$ significance level, three pairwise comparisons were statistically significant: Upbeat vs. Silence (adjusted $p = 0.0076$), Upbeat vs. Lo-Fi (adjusted $p = 0.0180$), and Upbeat vs. White Noise (adjusted $p = 0.0465$). All other pairs—Silence vs. Lo-Fi, Silence vs. White Noise, and Lo-Fi vs. White Noise—were not statistically significant, indicating that these three conditions are associated with equivalent mean WPM. The confidence interval plot in Figure 8 confirms this pattern: only the intervals for comparisons involving Upbeat music exclude zero.

These results indicate that Upbeat music is the sole auditory condition driving the significant ANOVA result. On average, participants typed approximately 4–5 fewer words per minute under Upbeat music compared to the other three conditions. Silence, Lo-Fi, and White Noise produced statistically indistinguishable typing speeds, suggesting that any of these environments can support equivalent typing performance.

Conclusion/Overall Results

We started our discussion with the question “does what you listen to while typing affect productivity through typing speed?”. Using data from 72 independent trials, we can draw the following conclusions. Silence, Lofi, and White noise all produced statistically equivalent typing speeds which means that the audio did not impact overall typing speed and productivity. The only audio condition that did alter productivity was upbeat music.

Upbeat music had consistently performed worse than the other conditions with 4 to 5 less words per trial. This is in the span of one minute, when spanned across hours of work, this can lead to multiple paragraphs of a difference. Typing seems to require a calm and smooth environment for the brain to comprehend and fire off motor skills sharply. The lyrics and rhythms in upbeat music appear to disrupt the cognitive attention that optimal typing requires. If the individual goal is to type as fast and productive as possible, there is evidence to suggest avoiding upbeat music.

Interestingly, Lofi music and White noise perform similarly to silence which provides a good option for those that do not want to work in silence.

This study has several limitations worth acknowledging. The sample consists of only three participants, all members of the same student group, which limits the generalisability of the findings to a broader population of typists with diverse backgrounds and typing habits. Although accuracy was recorded during data collection, it was not included as a response variable in this analysis; future work could examine both speed and accuracy jointly, or investigate whether auditory conditions that reduce speed also increase error rates. In addition, all participants used

the same online typing platform, and familiarity with the interface may have introduced a learning effect over the two-week collection period. The randomised condition order reduces but does not eliminate this concern. Finally, the study focused exclusively on short-burst, one-minute trials; effects may differ for extended, real-world typing tasks requiring sustained concentration.

References

- R Core Team. (2024). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>
- Wickham, H. (2016). ggplot2: Elegant graphics for data analysis. Springer-Verlag New York. <https://ggplot2.tidyverse.org>
- Pruim, R., Kaplan, D. T., & Horton, N. J. (2017). The mosaic package: Helping students to ‘think with data’ using R. The R Journal, 9(1), 77–02. <https://journal.r-project.org/archive/2017/RJ-2017-024/>
- Fox, J., & Weisberg, S. (2019). An R companion to applied regression (3rd ed.). Sage. <https://socialsciences.mcmaster.ca/jfox/Books/Companion/>
- Monkeytype. (2024). Monkeytype: A minimalist, customisable typing test. <https://monkeytype.com>

Appendix

Data imported from csv file as `typing_combined`, variables being “WPM”, “Tester”, and “Music”

```
library(ggplot2); library(mosaic); library(car); options(scipen = 999)
```

```
typing.df <- read.csv("data/typing_combined.csv"); typing.df$Music <-  
as.factor(typing.df$Music); typing.df$Tester <- as.factor(typing.df$Tester)
```

```
typing.aov <- aov(WPM ~ Music + Tester, data = typing.df)
```

```
e_terms <- residuals(typing.aov); fitted_vals <- fitted(typing.aov); typing.diag.df <-  
cbind(typing.df, e_terms, fitted_vals)
```

```
ggplot(data = typing.diag.df, aes(sample = e_terms)) + stat_qq(size = 2, col = "blue") +  
stat_qqline(col = "red") + ggtitle("Normal Probability Plot of the Residuals")
```

```
ggplot(data = typing.diag.df, aes(x = fitted_vals, y = e_terms)) + geom_point(size = 2, col =  
"blue") + xlab("Fitted Values") + ylab("Residuals") + geom_hline(yintercept = 0, linetype =  
"dashed", col = "red") + ggtitle("Residual Plot")
```

```
shapiro.test(e_terms); leveneTest(WPM ~ Music, data = typing.df); summary(typing.aov)
```

```
tukey.music <- TukeyHSD(typing.aov, "Music", conf.level = 0.95); tukey.music;  
plot(tukey.music, las = 1)
```